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A Python Library for Mathematical Finance

(Option Pricing and Sensitivities, Volatility Modelling, and Stochastic Processes)

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Contents

1 Option Pricing and Sensitivities 3
- 1.1 Mathematical Notation . . . 3
- 1.2 Option List . . . 3

European Vanilla Options (via the Black-Scholes Model) 5
- 1.3 European Vanilla Call Option . . . 5
- 1.4 European Vanilla Put Option . . . 5

European Vanilla Options (via the Heston Stochastic Volatility Model) 7
- 1.5 European Vanilla Call Option . . . 7
- 1.6 European Vanilla Put Option . . . 7
- 1.7 Sensitivities . . . 7
- 1.8 Parameter Calibration . . . 8

Binary Options (via the Black-Scholes Model) 9
- 1.9 Cash-or-Nothing Call . . . 9
- 1.10 Cash-or-Nothing Put . . . 9
- 1.11 Asset-or-Nothing Call . . . 10
- 1.12 Asset-or-Nothing Put . . . 10
- 1.13 Gap Call . . . 10
- 1.14 Gap Put . . . 11
- 1.15 Supershare . . . 11

Single Barrier Options (via the Reiner-Rubinstein Model) 12
- 1.16 Up-and-In Call . . . 12
- 1.17 Up-and-In Put . . . 12
- 1.18 Up-and-Out Call . . . 13
- 1.19 Up-and-Out Put . . . 13
- 1.20 Down-and-In Call . . . 13
- 1.21 Down-and-In Put . . . 13
- 1.22 Down-and-Out Call . . . 13
- 1.23 Down-and-Out Put . . . 13
- 1.24 Sensitivities . . . 13

Double Barrier Options (via the Kunitomo-Ikeda Model) 14
- 1.25 Boundary Curvature . . . 14
- 1.26 Double Knock-Out Call . . . 14
- 1.27 Double Knock-Out Put . . . 15
- 1.28 Double Knock-In Call . . . 15
- 1.29 Double Knock-In Put . . . 15
- 1.30 Sensitivities . . . 15

Barrier Binary Options (via the Reiner-Rubinstein Model) 16

Payout at Hit - Barrier Only 16
- 1.31 Up-and-In, Cash-at-Hit . . . 16
- 1.32 Down-and-In, Cash-at-Hit . . . 16
- 1.33 Up-and-In, Asset-at-Hit . . . 16
- 1.34 Down-and-In, Asset-at-Hit . . . 16
- 1.35 Sensitivities . . . 17

Payout at Maturity - Barrier Only 17
- 1.36 Up-and-In, Cash-or-Nothing . . . 17
- 1.37 Down-and-In, Cash-or-Nothing . . . 17
- 1.38 Up-and-In, Asset-or-Nothing . . . 17
- 1.39 Down-and-In, Asset-or-Nothing . . . 17
- 1.40 Up-and-Out, Cash-or-Nothing . . . 17
- 1.41 Down-and-Out, Cash-or-Nothing . . . 17

1.42 Up-and-Out, Asset-or-Nothing . . . 17
1.43 Down-and-Out, Asset-or-Nothing . . . 17
1.44 Sensitivities . . . 17

Payout at Maturity - Barrier and Strike **17**
1.45 Up-and-In Call, Cash-or-Nothing . . . 17
1.46 Down-and-In Call, Cash-or-Nothing . . . 18
1.47 Up-and-In Call, Asset-or-Nothing . . . 18
1.48 Down-and-In Call, Asset-or-Nothing . . . 18
1.49 Up-and-In Put, Cash-or-Nothing . . . 18
1.50 Down-and-In Put, Cash-or-Nothing . . . 18
1.51 Up-and-In Put, Asset-or-Nothing . . . 18
1.52 Down-and-In Put, Asset-or-Nothing . . . 18
1.53 Up-and-Out Call, Cash-or-Nothing . . . 18
1.54 Down-and-Out Call, Cash-or-Nothing . . . 18
1.55 Up-and-Out Call, Asset-or-Nothing . . . 19
1.56 Down-and-Out Call, Asset-or-Nothing . . . 19
1.57 Up-and-Out Put, Cash-or-Nothing . . . 19
1.58 Down-and-Out Put, Cash-or-Nothing . . . 19
1.59 Up-and-Out Put, Asset-or-Nothing . . . 19
1.60 Down-and-Out Put, Asset-or-Nothing . . . 19
1.61 Sensitivities . . . 19

Compound Options (via the Geske Model) **20**
1.62 Call-on-a-Call . . . 20
1.63 Call-on-a-Put . . . 20
1.64 Put-on-a-Call . . . 20
1.65 Put-on-a-Put . . . 20
1.66 Sensitivities . . . 20

Exchange Options (via the Margrabe Formula) **21**
1.67 Exchange Option . . . 21

Forward-Start Options (via the Rubinstein Formula) **22**
1.68 Forward-Start Standard Call . . . 22
1.69 Forward-Start Standard Put . . . 22

FX Options (via the Garman-Kohlhagen Model) **23**
1.70 European Call FX Option . . . 23
1.71 European Put FX Option . . . 23

Options on Futures (via the Black-76 Model) **24**
1.72 European Call Options on Futures . . . 24
1.73 European Put Options on Futures . . . 24

IN PROGRESS (Further Option Pricing Models) **25**

2 Volatility Modelling **26**

Implied Volatility (via the Black-Scholes Model) **26**

IN PROGRESS (Volatility Modelling Section) **27**

3 Stochastic Processes **28**

IN PROGRESS (Stochastic Processes Section) **28**

Option Pricing and Sensitivities

1.1 Mathematical Notation

Mathematical notation unless stated otherwise.

B	Predetermined fixed cash amount
δ_1, δ_2	Boundary curvature given by upper and lower barriers respectively
E	Strike price
F	Futures price
J	Rebate
κ	Speed of mean reversion
λ	Variance risk premium
Φ	Cumulative distribution function of a standard normal random variable
Φ_2	Cumulative distribution function of a standard bivariate normal random variable
q	Dividend yield
r	Risk-free interest rate
r_d	Domestic risk-free interest rate
r_f	Foreign risk-free interest rate
ρ	Correlation between underlying asset returns and variance shocks
S	Current underlying asset price
S_T	Terminal underlying asset price
σ	Underlying asset volatility
T	Time to maturity in years
θ	Long-run variance
v_0	Initial variance
X	Barrier price
X_d	Lower barrier price
X_u	Upper barrier price
ξ	Volatility of variance

1.2 Option List

A comprehensive list of all vanilla and exotic options priced, via the underlying Black-Scholes model (other than Heston), researched in this document.

Asset-or-Nothing Call	(*Binary*)
Asset-or-Nothing Put	(*Binary*)
Call-on-a-Call	(*Compound*)
Call-on-a-Put	(*Compound*)
Cash-or-Nothing Call	(*Binary*)
Cash-or-Nothing Put	(*Binary*)
Down-and-In, Asset-at-Hit	(*Barrier Binary: Payout at Hit*)
Down-and-In, Asset-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier Only*)
Down-and-In, Cash-at-Hit	(*Barrier Binary: Payout at Hit*)
Down-and-In, Cash-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier Only*)
Down-and-In Call, Asset-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)
Down-and-In Call, Cash-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)
Down-and-In Put, Asset-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)
Down-and-In Put, Cash-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)
Down-and-Out, Asset-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier Only*)
Down-and-Out, Cash-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier Only*)
Down-and-Out Call, Asset-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)
Down-and-Out Call, Cash-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)
Down-and-Out Put, Asset-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)
Down-and-Out Put, Cash-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)
Down-and-In Call	(*Single Barrier*)
Down-and-In Put	(*Single Barrier*)
Down-and-Out Call	(*Single Barrier*)
Down-and-Out Put	(*Single Barrier*)
Double Knock-In Call	(*Double Barrier*)
Double Knock-In Put	(*Double Barrier*)

Double Knock-Out Call	(*Double Barrier*)
Double Knock-Out Put	(*Double Barrier*)
European Call on Futures	(*Option on Futures*)
European FX Call	(*FX Option*)
European FX Put	(*FX Option*)
European Put on Futures	(*Option on Futures*)
European Vanilla Call	(*European: Black-Scholes*)
European Vanilla Call	(*European: Heston*)
European Vanilla Put	(*European: Black-Scholes*)
European Vanilla Put	(*European: Heston*)
Exchange	(*Exchange: Margrabe*)
Forward-Start Standard Call	(*Forward-Start*)
Forward-Start Standard Put	(*Forward-Start*)
Gap Call	(*Binary*)
Gap Put	(*Binary*)
Put-on-a-Call	(*Compound*)
Put-on-a-Put	(*Compound*)
Supershare	(*Binary*)
Up-and-In, Asset-at-Hit	(*Barrier Binary: Payout at Hit*)
Up-and-In, Asset-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier Only*)
Up-and-In, Cash-at-Hit	(*Barrier Binary: Payout at Hit*)
Up-and-In, Cash-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier Only*)
Up-and-In Call	(*Single Barrier*)
Up-and-In Call, Asset-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)
Up-and-In Call, Cash-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)
Up-and-In Put	(*Single Barrier*)
Up-and-In Put, Asset-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)
Up-and-In Put, Cash-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)
Up-and-Out, Asset-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier Only*)
Up-and-Out, Cash-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier Only*)
Up-and-Out Call	(*Single Barrier*)
Up-and-Out Call, Asset-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)
Up-and-Out Call, Cash-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)
Up-and-Out Put	(*Single Barrier*)
Up-and-Out Put, Asset-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)
Up-and-Out Put, Cash-or-Nothing	(*Barrier Binary: Payout at Maturity – Barrier and Strike*)

European Vanilla Options (via the Black-Scholes Model)

Define:

$$d_1 = \frac{\ln\left(\frac{S}{E}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

where S is the price of the underlying asset, E is the strike price, T is time to maturity in years, r is the risk-free interest rate, q is the dividend yield, and σ is the volatility of the underlying asset.

1.3 European Vanilla Call Option

A financial option is a financial derivative which represents a contract between the holder and the writer. The holder has the right, but not the obligation, to buy or sell an underlying asset at a predetermined price, known as the strike price, within a specific time frame. The holder pays a fee, known as a premium, to the writer for this right.

A European vanilla call option is a contract where at a certain time in the future, known as the expiration date, the holder of the option may choose to purchase the underlying asset at a certain price, known as the strike price. The term 'European' means that the financial option can only be exercised at expiry.

To find the closed-form analytical solution to the call price, we solve the Black-Scholes equation:

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + (r - q)S\frac{\partial C}{\partial S} - rC = 0.$$

1.3.1 Price

The price of a call option, C, is:

$$C = Se^{-qT}\Phi\left(d_1\right) - Ee^{-rT}\Phi\left(d_2\right), \tag{1}$$

where $\Phi(x)$ denotes the cumulative distribution function (CDF) of the standard normal distribution.

1.3.2 Sensitivities

Let $C = V$, and introduce the Greeks (Delta, Gamma, Rho, Theta, Vega respectively):

$$\Delta = \frac{\partial V}{\partial S}, \qquad \Gamma = \frac{\partial^2 V}{\partial S^2} = \frac{\partial \Delta}{\partial S}, \qquad \mathrm{P} = \frac{\partial V}{\partial r}, \qquad \Theta = \frac{\partial V}{\partial t}, \qquad \mathcal{V} = \frac{\partial V}{\partial \sigma}.$$

Delta (Δ) is how the option price will change with a change in the underlying asset price, Gamma (Γ) is the rate of change of Delta, Rho (P) is how the option price will change with a change in the risk-free interest rate, Theta (Θ) is how the option price will change as time passes, and Vega $(\mathcal{V})$ is how the option price will change with a change in the underlying asset volatility. Thus:

$$\Delta = e^{-qT}\Phi\left(d_1\right), \qquad \Gamma = e^{-qT}\frac{\phi\left(d_1\right)}{S\sigma\sqrt{T}}, \qquad \mathrm{P} = Ee^{-rT}T\Phi\left(d_2\right), \tag{2}$$
$$\Theta = -Se^{-qT}\phi(d_1)\frac{\sigma}{2\sqrt{T}} + qSe^{-qT}\Phi(d_1) - rEe^{-rT}\Phi(d_2), \qquad \mathcal{V} = Se^{-qT}\sqrt{T}\,\phi\left(d_1\right),$$

where $\phi(x) = \Phi'(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}$ is the probability density function (PDF) of the standard normal distribution.

1.4 European Vanilla Put Option

A European vanilla put option is where the holder of the option may choose to sell the underlying asset at the strike price.

To find the closed-form analytical solution to the put price, we solve the Black-Scholes equation:

$$\frac{\partial P}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 P}{\partial S^2} + (r - q)S\frac{\partial P}{\partial S} - rP = 0.$$

1.4.1 Price

The price of a call option, P, is:

$$P = Ee^{-rT}\Phi\left(-d_2\right) - Se^{-qT}\Phi\left(-d_1\right). \tag{3}$$

1.4.2 Sensitivities

The Greeks are:

$$\Delta = e^{-qT}\left(\Phi\left(d_1\right) - 1\right), \qquad \Gamma = e^{-qT}\frac{\phi\left(d_1\right)}{S\sigma\sqrt{T}}, \qquad \mathrm{P} = -Ee^{-rT}T\Phi\left(-d_2\right), \tag{4}$$

$$\Theta = -Se^{-qT}\phi(d_1)\frac{\sigma}{2\sqrt{T}} - qSe^{-qT}\Phi(-d_1) + rEe^{-rT}\Phi(-d_2), \qquad \mathcal{V} = Se^{-qT}\sqrt{T}\,\phi\left(d_1\right).$$

European Vanilla Options (via the Heston Stochastic Volatility Model)

Heston's model assumes that the underlying asset, S, follows a diffusion with a stochastic variance, v, whilst the variance itself is modelled as a mean-reverting square-root process. Note that r is the risk-free interest rate, κ is the speed of mean reversion, θ is the long-run variance, ξ is the volatility of variance, ρ is the correlation between asset returns and variance shocks, and λ is a parameter known as the variance risk premium.

The analytical solution to the Heston model is not easily implemented due to numerical issues. We can utilise an implementation that follows a single integral formulation closely related to Lewis' approximation formulae [2001]. We make sure to use the little Heston trap [2007], popularised by Gatheral's *'The Volatility Smile'*.

1.5 European Vanilla Call Option

Using standard arbitrage arguments, under the Heston model, we arrive at Garman's partial differential equation:

$$\frac{\partial C}{\partial t} + \frac{1}{2} v S^2 \frac{\partial^2 C}{\partial S^2} + \rho \xi v S \frac{\partial^2 C}{\partial v \partial S} + \frac{1}{2} \xi^2 v \frac{\partial^2 C}{\partial v^2} + rS \frac{\partial C}{\partial S} + \left[\kappa(\theta - v) - \lambda v\right] \frac{\partial C}{\partial v} - rC = 0.$$

Let $X = \ln(S)$ and $\tau = T - t$, in most cases $\tau = T$, since we take $t = 0$ as the current point in time, but since τ looks the part, we'll use it for the Heston model. Thus, the closed-form analytical solution to the fair value call price, C, is:

$$C\left(S, E, v_0, \tau\right) = \frac{1}{2}\left(Se^{-q\tau} - Ee^{-r\tau}\right) + \frac{1}{\pi} \int_0^\infty \Re\left[\frac{e^{-r\tau}\left(\varphi\left(\phi - i\right) - E\varphi\left(\phi\right)\right)}{i\phi E^{i\phi}}\right] d\phi, \tag{5}$$

where:

$$\varphi\left(X, E, v_0, \tau; \phi\right) = e^{(r-q)\phi i \tau} S^{i\phi} \left(\frac{1 - ge^{-d\tau}}{1 - g}\right)^{-\frac{2a}{\xi^2}} \exp\left[\frac{a\tau}{\xi^2}\left(b - \rho\xi\phi i - d\right) + \frac{v_0}{\xi^2}\left(b - \rho\xi\phi i - d\right)\left(\frac{1 - e^{-d\tau}}{1 - ge^{-d\tau}}\right)\right],$$

where:

$$g = \frac{b - \rho\xi\phi i - d}{b - \rho\xi\phi i + d}, \qquad d = \sqrt{\left(\rho\xi\phi i - b\right)^2 + \xi^2\left(\phi i + \phi^2\right)}, \qquad a = \kappa\theta, \qquad b = \kappa + \lambda.$$

1.6 European Vanilla Put Option

Under Put-Call parity:

$$C - P = Se^{-q\tau} - Ee^{-r\tau}.$$

Thus, the closed-form analytical solution to the fair value put price, P, is:

$$P\left(S, E, v_0, \tau\right) = \frac{1}{2}\left(Ee^{-r\tau} - Se^{-q\tau}\right) + \frac{1}{\pi} \int_0^\infty \Re\left[\frac{e^{-r\tau}\left(\varphi\left(\phi - i\right) - E\varphi\left(\phi\right)\right)}{i\phi E^{i\phi}}\right] d\phi. \tag{6}$$

1.7 Sensitivities

To find the sensitivities, we estimate via finite difference methods.

Using the central difference approximation:

$$\begin{aligned} \Delta &\approx \frac{V(S+h) - V(S-h)}{2h}, \qquad \mathrm{P} \approx \frac{V(r+h) - V(r-h)}{2h}, \\ \Theta &\approx -\frac{V(\tau+h) - C(\tau-h)}{2h}, \qquad \mathcal{V} \approx 2\sqrt{v_0}\left(\frac{V(v_0+h) - V(v_0-h)}{2h}\right). \end{aligned} \tag{7}$$

Using the second-order central difference approximation:

$$\Gamma \approx \frac{V(S+h) - 2V(S) + V(S-h)}{h^2}. \tag{8}$$

Using a mixture of first and second-order central difference approximations:

$$\mathrm{Volga} \approx 2\left(\frac{V(v_0+h) - V(v_0-h)}{2h}\right) + 4\sqrt{v_0}\left(\frac{V(v_0+h) - 2V(v_0) + V(v_0-h)}{h^2}\right). \tag{9}$$

Using a mixed derivative central difference approximation:

$$\mathrm{Vanna} \approx 2\sqrt{v_0}\left(\frac{V(S+h, v_0+h) - V(S+h, v_0-h) - V(S-h, v_0+h) + V(S-h, v_0-h)}{4h^2}\right). \tag{10}$$

1.8 Parameter Calibration

If there are unknown parameters, we must calibrate. Let $\Theta = (v_0, \kappa, \theta, \xi, \rho, \lambda)$. The aim of the calibration is to minimise the ordinary least squared errors:

$$SqErr(\Theta) = \sum_{i=1}^{N} \sum_{j=1}^{M} w_{ij} \left[V_{Market}\left(X_i, \tau_j\right) - V_{Heston}\left(S_\tau, X_i, \tau_j, r_j, \Theta\right) \right]^2 .$$

The optimisation objective function is:

$$\hat{\Theta} = \underset{\Theta \in U_\Theta}{\arg\min}\, SqErr\left(\Theta\right),$$

where we assume that the set of possible combinations of parameters, U_Θ, is compact and in the range for which a solution exists.

Binary Options (via the Black-Scholes Model)

Path-Independent binary options can be seen as extensions of the Black-Scholes-Merton model (Black-Scholes [1973] and Merton [1973]) where the binary pay-off structures differ from those of European vanilla options. In '*Unscrambling the Binary Code*' by Reiner and Rubinstein [1991b] there are developed valuation formulae for a wide range of path-independent and path-dependent binary options.

A cash-or-nothing option pays a fixed cash amount if the underlying asset matures above the strike price for a call, or below the strike price for a put; otherwise, it expires worthless.

An asset-or-nothing option pays one unit of the underlying asset if the option finishes in-the-money; otherwise, it expires worthless.

A gap option uses one strike price to determine whether the option is exercised and a second strike price to determine the size of the pay-off.

A supershare option pays one unit of the underlying asset if the terminal asset price finishes within a specified range bounded by two strike prices; otherwise, it expires worthless. The supershare concept was originally introduced by Nils Hakansson [1976], who proposed it as a financial instrument that pays only when the return of an underlying asset falls within a specified range of outcomes, allowing investors to construct a wide variety of pay-off structures from combinations of contracts.

1.9 Cash-or-Nothing Call

Let B represent a constant sum of money, that is, a predetermined fixed cash amount. The pay-off for a cash-or-nothing call option is $\Pi = B \cdot \mathbf{1}_{\{S_T > E\}} = BH(S_T - E)$, where H represents the Heaviside step function, that is, $H(x) = \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases}$.

1.9.1 Price

Thus, without redefining d_1 and d_2 from the original Black-Scholes model, seen in sections (1.3) and (1.4), the fair value of a cash-or-nothing call option is:

$$C = Be^{-rT}\Phi(d_2). \tag{11}$$

1.9.2 Sensitivities

$$\Delta = \frac{Be^{-rT}\phi(d_2)}{S\sigma\sqrt{T}}, \qquad \Gamma = -\frac{Be^{-rT}\phi(d_2)d_1}{S^2\sigma^2 T}, \qquad \mathrm{P} = Be^{-rT}\left(\frac{\sqrt{T}}{\sigma}\phi(d_2) - T\Phi(d_2)\right), \tag{12}$$
$$\Theta = Be^{-rT}\left(\frac{\phi(d_2)}{2\sigma T^{\frac{3}{2}}}\left(\ln\left(\frac{S}{E}\right) - \left(r - q - \frac{\sigma^2}{2}\right)T\right) + r\Phi(d_2)\right), \qquad \mathcal{V} = -\frac{Be^{-rT}}{\sigma}d_1\phi(d_2).$$

1.10 Cash-or-Nothing Put

1.10.1 Price

The fair value of a cash-or-nothing put option is:

$$P = Be^{-rT}\Phi(-d_2). \tag{13}$$

1.10.2 Sensitivities

$$\Delta = -\frac{Be^{-rT}\phi(d_2)}{S\sigma\sqrt{T}}, \qquad \Gamma = \frac{Be^{-rT}\phi(d_2)d_1}{S^2\sigma^2 T}, \qquad \mathrm{P} = -Be^{-rT}\left(\frac{\sqrt{T}}{\sigma}\phi(d_2) + T\Phi(-d_2)\right), \tag{14}$$
$$\Theta = Be^{-rT}\left(-\frac{\phi(d_2)}{2\sigma T^{\frac{3}{2}}}\left(\ln\left(\frac{S}{E}\right) - \left(r - q - \frac{\sigma^2}{2}\right)T\right) + r\Phi(-d_2)\right), \qquad \mathcal{V} = \frac{Be^{-rT}}{\sigma}d_1\phi(d_2).$$

1.11 Asset-or-Nothing Call

The pay-off for an asset-or-nothing call is $\Pi = S_T \cdot \mathbf{1}_{\{S_T > E\}} = S_T H(S_T - E)$.

1.11.1 Price

Thus, the fair value of an asset-or-nothing call option, where S represents the current underlying asset price, is:

$$C = Se^{-qT}\Phi(d_1). \tag{15}$$

1.11.2 Sensitivities

$$\Delta = \frac{e^{-qT}\phi(d_1)}{\sigma\sqrt{T}} + e^{-qT}\Phi(d_1), \qquad \Gamma = -\frac{e^{-qT}\phi(d_2)d_1}{S\sigma^2 T}, \qquad \mathrm{P} = \frac{Se^{-qT}\sqrt{T}}{\sigma}\phi(d_1), \tag{16}$$
$$\Theta = Se^{-qT}\left(\frac{\phi(d_1)}{2\sigma T^{\frac{3}{2}}}\left(\ln\left(\frac{S}{E}\right) - \left(r - q + \frac{\sigma^2}{2}\right)T\right) + q\Phi(d_1)\right), \qquad \mathcal{V} = -\frac{Se^{-qT}}{\sigma}d_2\phi(d_1).$$

1.12 Asset-or-Nothing Put

1.12.1 Price

The fair value of an asset-or-nothing call option is:

$$P = Se^{-qT}\Phi(-d_1). \tag{17}$$

1.12.2 Sensitivities

$$\Delta = -\frac{e^{-qT}\phi(d_1)}{\sigma\sqrt{T}} + e^{-qT}\Phi(-d_1), \qquad \Gamma = \frac{e^{-qT}\phi(d_2)d_1}{S\sigma^2 T}, \qquad \mathrm{P} = -\frac{Se^{-qT}\sqrt{T}}{\sigma}\phi(d_1), \tag{18}$$
$$\Theta = Se^{-qT}\left(-\frac{\phi(d_1)}{2\sigma T^{\frac{3}{2}}}\left(\ln\left(\frac{S}{E}\right) - \left(r - q + \frac{\sigma^2}{2}\right)T\right) + q\Phi(-d_1)\right), \qquad \mathcal{V} = \frac{Se^{-qT}}{\sigma}d_2\phi(d_1).$$

1.13 Gap Call

Gap options are slightly more complex than standard options. These options highlight the dual role played by the strike price, E_1 in a standard option; E_1 not only determines whether the option finishes in or out of the money but also the size of the resulting pay-off, E_2. The gap is defined as $E_2 - E_1$. Do note that E_1 is the exercise strike, whilst E_2 is the pay-off strike. Positive gap calls will sell for less than standard calls, whilst positive gap puts will be worth more than their standard counterparts. The value of a gap option may be derived by subtracting a cash-or-nothing option from an asset-or-nothing option. The pay-off is $\Pi = \max(S_T - E_2, 0)$, depending if $S_T > E_1$ or $S_T \leq E_1$ respectively. Do note d_1 and d_2 do not change in definition, just with E_1.

1.13.1 Price

Thus, the fair value of a gap call option is:

$$C = Se^{-qT}\Phi(d_1) - E_2 e^{-rT}\Phi(d_2). \tag{19}$$

This is almost identical to the Black-Scholes formula for a European vanilla option.

1.13.2 Sensitivities

Let asset-or-nothing be AON and cash-or-nothing be CON, thus, since a gap option can be decomposed as:

$$V = V_{AON} - E_2 V_{\text{unit } CON},$$

then each Greek can be decomposed similarly. That is:

$$\Delta = \Delta_{AON} - E_2\Delta_{\text{unit } CON}, \qquad \Gamma = \Gamma_{AON} - E_2\Gamma_{\text{unit } CON}, \qquad \mathrm{P} = \mathrm{P}_{AON} - E_2\mathrm{P}_{\text{unit } CON}, \tag{20}$$
$$\Theta = \Theta_{AON} - E_2\Theta_{\text{unit } CON}, \qquad \mathcal{V} = \mathcal{V}_{AON} - E_2\mathcal{V}_{\text{unit } CON}.$$

Note that we take the call Greeks of each.

1.14 Gap Put

1.14.1 Price

The fair value of a gap put option is:

$$P = E_2 e^{-rT}\Phi(-d_2) - Se^{-qT}\Phi(-d_1). \tag{21}$$

1.14.2 Sensitivities

Take the same equations as (20) but use the put Greeks of each.

1.15 Supershare

Note that supershares share the same pay-off as ($\frac{1}{E_d}$) bullish vertical spreads of asset-or-nothing calls. Take the two strikes with the lower bound, E_d, and the upper bound, E_u. If $d_1 = d_1(E)$, then $d_1(E_d)$ is d_1 with the lower strike and $d_1(E_u)$ is d_1 with the upper strike.

1.15.1 Price

Therefore, the fair value of a supershare is:

$$V = \frac{Se^{-qT}}{E_d}\left(\Phi\left(d_1\left(E_d\right)\right) - \Phi\left(d_1\left(E_u\right)\right)\right). \tag{22}$$

1.15.2 Sensitivities

Since a supershare option can be viewed as:

$$V = \frac{1}{E_d}\left(C_{AON}(E_d) - C_{AON}(E_u)\right),$$

then each Greek can be viewed similarly. That is:

$$\begin{aligned}
\Delta &= \frac{1}{E_d}\left(\Delta_{AON}(E_d) - \Delta_{AON}(E_u)\right), \qquad \Gamma = \frac{1}{E_d}\left(\Gamma_{AON}(E_d) - \Gamma_{AON}(E_u)\right), \\
\mathrm{P} &= \frac{1}{E_d}\left(\mathrm{P}_{AON}(E_d) - \mathrm{P}_{AON}(E_u)\right), \qquad \Theta = \frac{1}{E_d}\left(\Theta_{AON}(E_d) - \Theta_{AON}(E_u)\right), \\
\mathcal{V} &= \frac{1}{E_d}\left(\mathcal{V}_{AON}(E_d) - \mathcal{V}_{AON}(E_u)\right).
\end{aligned} \tag{23}$$

Note that we take the call Greeks of each.

Single Barrier Options (via the Reiner-Rubinstein Model)

To derive closed-form analytical solutions to barrier style options, we can use the Reiner-Rubinstein model [1991a], which utilises the same assumption as Black-Scholes. Ultimately, it can be thought of as a combination to the Black-Scholes and the reflection principle. The underlying still follows a geometric Brownian motion.

A barrier option is an option which has an additional constraint such that it becomes active, or inactive, if the underlying asset, S, passes through a barrier, X. Only active options can be exercised. For an in-barrier, the option cannot be exercised unless the underlying passes through the barrier from either above or below. For an out-barrier, the option becomes inactive, hence worthless, if it passes through the barrier either from above or below. The In-Out parity states the fair-value of a knock-in and a knock-out barrier option equal a vanilla option.

We will examine the closed-form analytical solutions to European style barrier options.

Define:

$$A_1 = \phi S e^{-qT}\Phi(\phi x_1) - \phi E e^{-rT}\Phi\left(\phi x_1 - \phi\sigma\sqrt{T}\right),$$

$$A_2 = \phi S e^{-qT}\Phi(\phi x_1) - \phi E e^{-rT}\Phi\left(\phi x_2 - \phi\sigma\sqrt{T}\right),$$

$$A_3 = \phi\left(\frac{X}{S}\right)^{2(\mu+1)} S e^{-qT}\Phi(\eta x_3) - \phi\left(\frac{X}{S}\right)^{2\mu} E e^{-rT}\Phi\left(\eta x_3 - \eta\sigma\sqrt{T}\right),$$

$$A_4 = \phi\left(\frac{X}{S}\right)^{2(\mu+1)} S e^{-qT}\Phi(\eta x_4) - \phi\left(\frac{X}{S}\right)^{2\mu} E e^{-rT}\Phi\left(\eta x_4 - \eta\sigma\sqrt{T}\right),$$

$$A_5 = J e^{-rT}\left(\Phi\left(\eta x_2 - \eta\sigma\sqrt{T}\right) - \left(\frac{X}{S}\right)^{2\mu}\Phi\left(\eta x_4 - \eta\sigma\sqrt{T}\right)\right),$$

$$A_6 = J\left(\left(\frac{X}{S}\right)^{\mu+\lambda}\Phi\left(\eta x_5\right) + \left(\frac{X}{S}\right)^{\mu-\lambda}\Phi\left(\eta x_5 - 2\eta\lambda\sigma\sqrt{T}\right)\right),$$

where:

$$\phi = \begin{cases} 1, & \text{call} \\ -1, & \text{put} \end{cases}, \qquad \eta = \begin{cases} 1, & \text{down} \\ -1, & \text{up} \end{cases},$$

$$\mu = \frac{r - q - \frac{\sigma^2}{2}}{\sigma^2}, \qquad \lambda = \sqrt{\mu^2 + \frac{2r}{\sigma^2}},$$

$$x_1 = \frac{\ln\left(\frac{S}{E}\right)}{\sigma\sqrt{T}} + (\mu+1)\sigma\sqrt{T}, \qquad x_2 = \frac{\ln\left(\frac{S}{X}\right)}{\sigma\sqrt{T}} + (\mu+1)\sigma\sqrt{T},$$

$$x_3 = \frac{\ln\left(\frac{X^2}{SE}\right)}{\sigma\sqrt{T}} + (\mu+1)\sigma\sqrt{T}, \qquad x_4 = \frac{\ln\left(\frac{X}{S}\right)}{\sigma\sqrt{T}} + (\mu+1)\sigma\sqrt{T},$$

$$x_5 = \frac{\ln\left(\frac{X}{S}\right)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}.$$

If there is a rebate feature, J, we may be eligible for a cash amount payoff, increasing the fair value of said options. For all knock-in options, we use the rebate, J, in A_5. For all knock-out options, we use the rebate, J, in A_6.

1.16 Up-and-In Call

If $E \leq X$:

$$C = A_2 - A_3 + A_4 + A_5. \tag{24}$$

If $E > X$:

$$C = A_1 + A_5. \tag{25}$$

1.17 Up-and-In Put

If $E \leq X$:

$$P = A_3 + A_5. \tag{26}$$

If $E > X$:

$$P = A_1 - A_2 + A_4 + A_5. \tag{27}$$

1.18 Up-and-Out Call

If $E \leq X$:

$$C = A_1 - A_2 + A_3 - A_4 + A_6. \tag{28}$$

If $E > X$:

$$C = A_6. \tag{29}$$

1.19 Up-and-Out Put

If $E \leq X$:

$$P = A_1 - A_3 + A_6. \tag{30}$$

If $E > X$:

$$P = A_2 - A_4 + A_6. \tag{31}$$

1.20 Down-and-In Call

If $E \leq X$:

$$C = A_1 - A_2 + A_4 + A_5. \tag{32}$$

If $E > X$:

$$C = A_3 + A_5. \tag{33}$$

1.21 Down-and-In Put

If $E \leq X$:

$$P = A_1 + A_5. \tag{34}$$

If $E > X$:

$$P = A_2 - A_3 + A_4 + A_5. \tag{35}$$

1.22 Down-and-Out Call

If $E \leq X$:

$$C = A_2 - A_4 + A_6. \tag{36}$$

If $E > X$:

$$C = A_1 - A_3 + A_6. \tag{37}$$

1.23 Down-and-Out Put

If $E \leq X$:

$$P = A_6. \tag{38}$$

If $E > X$:

$$P = A_1 - A_2 + A_3 - A_4 + A_6. \tag{39}$$

1.24 Sensitivities

The sensitivities can be solved numerically via finite difference methods. Recall equations (7)-(10) to use central difference approximations.

Double Barrier Options (via the Kunitomo-Ikeda Model)

Double barrier options are either knocked in or knocked out if the underlying crosses the lower barrier, X_d, or the upper barrier, X_u, before the option expires.

Define:

$$\mu = \frac{2(r-q)}{\sigma^2} + 1,$$

$$d_1 = \frac{\ln\left(\frac{SX_u^{2n}}{EX_d^{2n}}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \qquad d_2 = \frac{\ln\left(\frac{SX_u^{2n-1}}{X_d^{2n}}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}},$$

$$d_3 = \frac{\ln\left(\frac{X_d^{2n+2}}{SEX_u^{2n}}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \qquad d_4 = \frac{\ln\left(\frac{X_d^{2n+2}}{SX_u^{2n+1}}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}.$$

Thus, the fair value of a double knock-out call option is:

$$C = Se^{-qT}\sum_{n=-\infty}^{\infty}\left[\left(\frac{X_u^n}{X_d^n}\right)^{\mu}(\Phi(d_1) - \Phi(d_2)) - \left(\frac{X_d^{n+1}}{SX_u^n}\right)^{\mu}(\Phi(d_3) - \Phi(d_4))\right] - Ee^{-rT}\sum_{n=-\infty}^{\infty}\left[\left(\frac{X_u^n}{X_d^n}\right)^{\mu-2}\right.$$
$$\left.\left(\Phi\left(d_1 - \sigma\sqrt{T}\right) - \Phi\left(d_2 - \sigma\sqrt{T}\right)\right) - \left(\frac{X_d^{n+1}}{SX_u^n}\right)^{\mu-2}\left(\Phi\left(d_3 - \sigma\sqrt{T}\right) - \Phi\left(d_4 - \sigma\sqrt{T}\right)\right)\right].$$

Using In-Out parity, $C_{DKI} = C_{BS} - C_{DKO}$, where C_{BS} is the call price using the Black-Scholes model.

However, in Kunitomo-Ikeda's original paper [1992], they introduced options pricing with curved boundaries.

1.25 Boundary Curvature

Sometimes, boundaries are not always flat, we can introduce variables, δ_1, δ_2, to represent boundary curvature. If $\delta_1, \delta_2 = 0$, the boundaries are flat. For $\delta_1 < 0 < \delta_2$, the lower boundary exhibits exponential growth and the upper boundary exhibits exponential decay. For $\delta_1 > 0 > \delta_2$, we have a convex lower boundary and convex upper boundary.

1.26 Double Knock-Out Call

Redefine:

$$\mu_1 = \frac{2\left(r - q - \delta_2 - n(\delta_1 - \delta_2)\right)}{\sigma^2} + 1, \qquad \mu_2 = \frac{2n(\delta_1 - \delta_2)}{\sigma^2}$$

$$\mu_3 = \frac{2\left(r - q - \delta_2 + n(\delta_1 - \delta_2)\right)}{\sigma^2} + 1, \qquad F_1 = X_u e^{\delta_1 T}$$

$$d_1 = \frac{\ln\left(\frac{SX_u^{2n}}{EX_d^{2n}}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \qquad d_2 = \frac{\ln\left(\frac{SX_u^{2n}}{F_1X_d^{2n}}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}},$$

$$d_3 = \frac{\ln\left(\frac{X_d^{2n+2}}{SEX_u^{2n}}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \qquad d_4 = \frac{\ln\left(\frac{X_d^{2n+2}}{F_1SX_u^{2n}}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}.$$

Thus, the fair value of a double knock-out call option adjusting for boundary curvature is:

$$C = Se^{-qT}\sum_{n=-\infty}^{\infty}\left[\left(\frac{X_u^n}{X_d^n}\right)^{\mu_1}\left(\frac{X_d}{S}\right)^{\mu_2}(\Phi(d_1) - \Phi(d_2)) - \left(\frac{X_d^{n+1}}{SX_u^n}\right)^{\mu_3}(\Phi(d_3) - \Phi(d_4))\right] - Ee^{-rT}$$
$$\sum_{n=-\infty}^{\infty}\left[\left(\frac{X_u^n}{X_d^n}\right)^{\mu_1-2}\left(\frac{X_d}{S}\right)^{\mu_2}\left(\Phi\left(d_1 - \sigma\sqrt{T}\right) - \Phi\left(d_2 - \sigma\sqrt{T}\right)\right) - \left(\frac{X_d^{n+1}}{SX_u^n}\right)^{\mu_3-2}\right.$$
$$\left.\left(\Phi\left(d_3 - \sigma\sqrt{T}\right) - \Phi\left(d_4 - \sigma\sqrt{T}\right)\right)\right]. \quad (40)$$

1.27 Double Knock-Out Put

Define:

$$F_2 = X_d e^{\delta_2 T}$$

$$y_1 = \frac{\ln\left(\frac{SX_u^{2n}}{F_2 X_d^{2n}}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \qquad y_2 = \frac{\ln\left(\frac{SX_u^{2n}}{EX_d^{2n}}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}},$$

$$y_3 = \frac{\ln\left(\frac{X_d^{2n+2}}{F_2 S X_u^{2n}}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \qquad y_4 = \frac{\ln\left(\frac{X_d^{2n+2}}{SEX_u^{2n}}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}.$$

Thus, the fair value of a double knock-out put option adjusting for boundary curvature is:

$$P = Ee^{-rT}\sum_{n=-\infty}^{\infty}\left[\left(\frac{X_u^n}{X_d^n}\right)^{\mu_1-2}\left(\frac{X_d}{S}\right)^{\mu_2}\left(\Phi\left(y_1-\sigma\sqrt{T}\right)-\Phi\left(y_2-\sigma\sqrt{T}\right)\right)-\left(\frac{X_d^{n+1}}{SX_u^n}\right)^{\mu_3-2}\right.$$
$$\left.\left(\Phi\left(y_3-\sigma\sqrt{T}\right)-\Phi\left(y_4-\sigma\sqrt{T}\right)\right)\right]-Se^{-qT}\sum_{n=-\infty}^{\infty}\left[\left(\frac{X_u^n}{X_d^n}\right)^{\mu_1}\left(\frac{X_d}{S}\right)^{\mu_2}\right.$$
$$\left.(\Phi(y_1)-\Phi(y_2))-\left(\frac{X_d^{n+1}}{SX_u^n}\right)^{\mu_3}(\Phi(y_3)-\Phi(y_4))\right]. \quad (41)$$

1.28 Double Knock-In Call

Using In-Out parity, $C_{DKI} = C_{BS} - C_{DKO}$, where C_{BS} is the call price using the Black-Scholes model.

1.29 Double Knock-In Put

Similarly, $P_{DKI} = P_{BS} - P_{DKO}$, where P_{BS} is the put price using the Black-Scholes model.

1.30 Sensitivities

The sensitivities can be solved numerically via finite difference methods. Recall equations (7)-(10) to use central difference approximations.

Barrier Binary Options (via the Reiner-Rubinstein Model)

Barrier Binary options are a type of path-independent binary option. There are three major types of Barrier Binary options: payout at hit - barrier only, payout at maturity - barrier only, and payout at maturity - barrier and strike.

Define:

$$A_1 = Se^{-qT}\Phi(\phi x_1), \qquad A_2 = Se^{-qT}\Phi(\phi x_2),$$
$$A_3 = Se^{-qT}\left(\frac{X}{S}\right)^{2\lambda}\Phi\left(\eta x_3\right), \qquad A_4 = Se^{-qT}\left(\frac{X}{S}\right)^{2\lambda}\Phi\left(\eta x_4\right),$$
$$C_1 = Be^{-rT}\Phi\left(\phi x_1 - \phi\sigma\sqrt{T}\right), \qquad C_2 = Be^{-rT}\Phi\left(\phi x_2 - \phi\sigma\sqrt{T}\right),$$
$$C_3 = Be^{-rT}\left(\frac{X}{S}\right)^{2\lambda-2}\Phi\left(\eta x_3 - \eta\sigma\sqrt{T}\right), \qquad C_4 = Be^{-rT}\left(\frac{X}{S}\right)^{2\lambda-2}\Phi\left(\eta x_4 - \eta\sigma\sqrt{T}\right),$$
$$D = B\left[\left(\frac{X}{S}\right)^{a+b}\Phi\left(\eta x_5\right) + \left(\frac{X}{S}\right)^{a-b}\Phi\left(\eta x_5 - 2\eta b\sigma\sqrt{T}\right)\right]$$

where:

$$\phi = \begin{cases} 1, & \text{call} \\ -1, & \text{put} \end{cases}, \qquad \eta = \begin{cases} 1, & \text{down} \\ -1, & \text{up} \end{cases},$$
$$\mu = r - q - \frac{1}{2}\sigma^2, \qquad \lambda = 1 + \left(\frac{\mu}{\sigma^2}\right),$$
$$a = \frac{\mu}{\sigma^2}, \qquad b = \frac{\sqrt{\mu^2 + 2r\sigma^2}}{\sigma^2},$$
$$x_1 = \frac{\ln\left(\frac{S}{E}\right)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}, \qquad x_2 = \frac{\ln\left(\frac{S}{X}\right)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}.$$
$$x_3 = \frac{\ln\left(\frac{X^2}{SE}\right)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}, \qquad x_4 = \frac{\ln\left(\frac{X}{S}\right)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T},$$
$$x_5 = \frac{\ln\left(\frac{X}{S}\right)}{\sigma\sqrt{T}} + b\sigma\sqrt{T}.$$

Note that $x_1 = d_1$, and the rest are quite similarly defined.

Payout at Hit - Barrier Only

These are for in barrier options only. Denote the fair value of each barrier binary option by V.

1.31 Up-and-In, Cash-at-Hit

$$V = D. \tag{42}$$

1.32 Down-and-In, Cash-at-Hit

$$V = D. \tag{43}$$

1.33 Up-and-In, Asset-at-Hit

Let $B = X$, *i.e.*, the predetermined fixed cash amount is equal to the barrier price.

$$V = D. \tag{44}$$

1.34 Down-and-In, Asset-at-Hit

Similarly, let $B = X$.

$$V = D. \tag{45}$$

1.35 Sensitivities

The sensitivities can be solved numerically via finite difference methods. Recall equations (7)-(10) to use central difference approximations.

Payout at Maturity - Barrier Only

Since we do not have the usual call or put. We define ϕ slightly differently. Thus, for (1.36) - (1.43):

$$\phi = \begin{cases} \eta, & \text{out} \\ -\eta, & \text{in} \end{cases}.$$

1.36 Up-and-In, Cash-or-Nothing

$$V = C_2 + C_4. \tag{46}$$

1.37 Down-and-In, Cash-or-Nothing

$$V = C_2 + C_4. \tag{47}$$

1.38 Up-and-In, Asset-or-Nothing

$$V = A_2 + A_4. \tag{48}$$

1.39 Down-and-In, Asset-or-Nothing

$$V = A_2 + A_4. \tag{49}$$

1.40 Up-and-Out, Cash-or-Nothing

$$V = C_2 - C_4. \tag{50}$$

1.41 Down-and-Out, Cash-or-Nothing

$$V = C_2 - C_4. \tag{51}$$

1.42 Up-and-Out, Asset-or-Nothing

$$V = A_2 - A_4. \tag{52}$$

1.43 Down-and-Out, Asset-or-Nothing

$$V = A_2 - A_4. \tag{53}$$

1.44 Sensitivities

The sensitivities can be solved numerically via finite difference methods. Recall equations (7)-(10) to use central difference approximations.

Payout at Maturity - Barrier and Strike

1.45 Up-and-In Call, Cash-or-Nothing

If $E \leq X$:

$$C = C_2 - C_3 + C_4. \tag{54}$$

If $E > X$:

$$C = C_1. \tag{55}$$

1.46 Down-and-In Call, Cash-or-Nothing

If $E \leq X$:

$$C = C_1 - C_2 + C_4. \tag{56}$$

If $E > X$:

$$C = C_3. \tag{57}$$

1.47 Up-and-In Call, Asset-or-Nothing

If $E \leq X$:

$$C = A_2 - A_3 + A_4. \tag{58}$$

If $E > X$:

$$C = A_1. \tag{59}$$

1.48 Down-and-In Call, Asset-or-Nothing

If $E \leq X$:

$$C = A_1 - A_2 + A_4. \tag{60}$$

If $E > X$:

$$C = A_3. \tag{61}$$

1.49 Up-and-In Put, Cash-or-Nothing

If $E \leq X$:

$$P = C_3. \tag{62}$$

If $E > X$:

$$P = C_1 - C_2 + C_4. \tag{63}$$

1.50 Down-and-In Put, Cash-or-Nothing

If $E \leq X$:

$$P = C_1. \tag{64}$$

If $E > X$:

$$P = C_2 - C_3 + C_4. \tag{65}$$

1.51 Up-and-In Put, Asset-or-Nothing

If $E \leq X$:

$$P = A_3. \tag{66}$$

If $E > X$:

$$P = A_1 - A_2 + A_3. \tag{67}$$

1.52 Down-and-In Put, Asset-or-Nothing

If $E \leq X$:

$$P = A_1. \tag{68}$$

If $E > X$:

$$P = A_2 - A_3 + A_4. \tag{69}$$

1.53 Up-and-Out Call, Cash-or-Nothing

If $E \leq X$:

$$C = C_1 - C_2 + C_3 - C_4. \tag{70}$$

If $E > X$:

$$C = 0. \tag{71}$$

1.54 Down-and-Out Call, Cash-or-Nothing

If $E \leq X$:

$$C = C_2 - C_4. \tag{72}$$

If $E > X$:

$$C = C_1 - C_3. \tag{73}$$

1.55 Up-and-Out Call, Asset-or-Nothing

If $E \leq X$:

$$C = A_1 - A_2 + A_3 - A_4. \tag{74}$$

If $E > X$:

$$C = 0. \tag{75}$$

1.56 Down-and-Out Call, Asset-or-Nothing

If $E \leq X$:

$$C = A_2 - A_4. \tag{76}$$

If $E > X$:

$$C = A_1 - A_3. \tag{77}$$

1.57 Up-and-Out Put, Cash-or-Nothing

If $E \leq X$:

$$P = C_1 - C_3. \tag{78}$$

If $E > X$:

$$P = C_2 - C_4. \tag{79}$$

1.58 Down-and-Out Put, Cash-or-Nothing

If $E \leq X$:

$$P = 0. \tag{80}$$

If $E > X$:

$$P = C_1 - C_2 + C_3 - C_4. \tag{81}$$

1.59 Up-and-Out Put, Asset-or-Nothing

If $E \leq X$:

$$P = A_1 - A_3. \tag{82}$$

If $E > X$:

$$P = A_2 - A_4. \tag{83}$$

1.60 Down-and-Out Put, Asset-or-Nothing

If $E \leq X$:

$$P = 0. \tag{84}$$

If $E > X$:

$$P = A_1 - A_2 + A_3 - A_4. \tag{85}$$

1.61 Sensitivities

The sensitivities can be solved numerically via finite difference methods. Recall equations (7)-(10) to use central difference approximations.

Compound Options (via the Geske Model)

A compound option is an option on an option, *i.e.*, the underlying asset is an option. There are more complicated combinations that exist, but the most common involve European vanilla calls and puts, *e.g.*, a Put-on-a-Call is where the compound option is a put, and the underlying option is a call. The advantage of compound options is that they allow large leverage, known as the gearing effect, and typically cheaper than standard options. To evaluate the analytical solution, Geske published a paper in [1979]. Let E_1, T_1 denote the compound option strike and time to maturity, and E_2, T_2 denote the underlying option strike and time to maturity. Note that $T_2 > T_1$.

Define:

$$A_1 = \phi\eta S e^{-qT_2}\Phi_2(\phi\eta x_1, \eta y_1; \phi\rho),$$
$$A_2 = \phi\eta E_2 e^{-rT_2}\Phi_2(\phi\eta x_2, \eta y_2; \phi\rho),$$
$$A_3 = \phi E_1 e^{-rT_1}\Phi(\phi\eta x_2)$$

where:

$$\phi = \begin{cases} 1, & \text{compound option is a call} \\ -1, & \text{compound option is a put} \end{cases}, \qquad \eta = \begin{cases} 1, & \text{underlying option is a call} \\ -1, & \text{underlying option is a put} \end{cases}, \qquad \rho = \sqrt{\frac{T_1}{T_2}},$$

$$x_1 = \frac{\ln\left(\frac{S}{S^*}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T_1}{\sigma\sqrt{T_1}}, \qquad x_2 = x_1 - \sigma\sqrt{T_1},$$

$$y_1 = \frac{\ln\left(\frac{S}{E_2}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T_2}{\sigma\sqrt{T_2}}, \qquad y_2 = y_1 - \sigma\sqrt{T_2}.$$

Note that S^* represents the critical underlying asset price S_{T_1} at which the value of the underlying standard option $V(S_{T_1}, T_1)$ equals the derivative option strike price, E_2. S^* can be found numerically using Newton-Raphson. S^* solves:

$$\eta S^* e^{-q\tau}\Phi(\eta z_1) - \eta E_2 e^{-r\tau}\Phi(\eta z_2) - E_1 = 0$$

where:

$$\tau = T_2 - T_1, \qquad z_1 = \frac{\ln\left(\frac{S^*}{E_2}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)\tau}{\sigma\sqrt{\tau}}, \qquad z_2 = z_1 - \sigma\sqrt{\tau}.$$

1.62 Call-on-a-Call

The fair value of the compound option, C, is:

$$C = A_1 - A_2 - A_3. \tag{86}$$

1.63 Call-on-a-Put

$$C = A_1 - A_2 - A_3. \tag{87}$$

1.64 Put-on-a-Call

$$C = A_1 - A_2 - A_3. \tag{88}$$

1.65 Put-on-a-Put

$$C = A_1 - A_2 - A_3. \tag{89}$$

1.66 Sensitivities

Note that Theta, Θ, the derivation of the option value with respect to time must take into account the passage of time as it affects the remaining time to maturity of the compound option T_1, and the remaining time to maturity of the underlying option, T_2. Thus:

$$\Theta = -\left(\left(\frac{\partial C}{\partial T_1}\right) + \left(\frac{\partial C}{\partial T_2}\right)\right).$$

The sensitivities can be solved numerically via finite difference methods. Recall equations (7)-(10) to use central difference approximations.

Exchange Options (via the Margrabe Formula)

An exchange option gives its owner the right, but not the obligation, to exchange one asset for another at a specified future date. Exchange options can either be interpreted as a call on asset 2 with a strike price equal to the future value of asset 1, or a put on asset 1 with a striking price equal to the future value of asset 2. Let ρ be the correlation coefficient of the natural logarithms of one plus their rates of returns, Σ be the combined relative volatility between the two underlying assets, S_1, S_2, q_1, q_2, and σ_1, σ_2 represent the underlying asset price, dividend yield, and underlying asset volatility for asset 1 and asset 2, respectively.

Redefine for section (1.67), such that:

$$d_1 = \frac{\ln\left(\frac{x_2}{x_1}\right) + \frac{1}{2}\Sigma^2 T}{\Sigma\sqrt{T}}, \qquad d_2 = d_1 - \Sigma\sqrt{T},$$

where:

$$x_1 = S_1 e^{-q_1 T}, \qquad x_2 = S_2 e^{-q_2 T},$$
$$\Sigma^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2.$$

1.67 Exchange Option

1.67.1 Price

Let asset 2 be the asset received and asset 1 be surrendered. Thus, the fair value of an exchange option is:

$$V = x_2\Phi\left(d_1\right) - x_1\Phi\left(d_2\right). \tag{90}$$

1.67.2 Sensitivities

When introducing the Greeks for exchange options, we might want to re-visualise, let Γ_{12} represent the Cross-Gamma and the correlation sensitivity, χ, also known as the Correlation Vega/Delta or correlation risk:

$$\Delta_1 = \frac{\partial V}{\partial S_1}, \qquad \Delta_2 = \frac{\partial V}{\partial S_2},$$
$$\Gamma_1 = \frac{\partial^2 V}{\partial S_1^2}, \qquad \Gamma_2 = \frac{\partial^2 V}{\partial S_2^2}, \qquad \Gamma_{12} = \frac{\partial^2 V}{\partial S_1 \partial S_2},$$
$$\chi = \frac{\partial V}{\partial \rho}, \qquad \Theta = \frac{\partial V}{\partial t}, \qquad \mathcal{V} = \frac{\partial V}{\partial \Sigma}.$$

Thus:

$$\Delta_1 = e^{-q_1 T}\Phi\left(d_2\right), \qquad \Delta_2 = e^{-q_2 T}\Phi\left(d_1\right),$$
$$\Gamma_1 = e^{-q_1 T}\frac{\phi\left(d_2\right)}{S_2\Sigma\sqrt{T}},, \qquad \Gamma_2 = e^{-q_2 T}\frac{\phi\left(d_1\right)}{S_2\Sigma\sqrt{T}}, \qquad \Gamma_{12} = -e^{-(q_1+q_2)T}\frac{\phi\left(d_1\right)}{S_1\Sigma\sqrt{T}}, \tag{91}$$
$$\chi = -x_2\left(\frac{\sigma_1\sigma_2}{\Sigma}\right)\sqrt{T}\,\phi(d_1), \qquad \mathcal{V} = x_2\sqrt{T}\,\phi\left(d_1\right),$$
$$\Theta = -x_2\left(\frac{\Sigma}{2\sqrt{T}}\right)\phi(d_1) + q_2 x_2\Phi(d_1) - q_1 x_1\Phi(d_2).$$

Note that in the Cross-Gamma, you could define using d_2 and S_2 instead.

Forward-Start Options (via the Rubinstein Formula)

A forward-start option is a standard European option that is agreed upon in the present but only becomes active at a pre-specified future date, t, which is known as the forward-start or grant date. The strike price, E, is set equal to the current underlying asset price, $S_t = S(t)$. The holder therefore has the right to receive an option in the future, rather than receiving the option immediately. Employee incentive stock options could be seen as a form of forward-start options.

The valuation of forward-start options exploits the fact that the pay-off can often be expressed in terms of the asset return over a time, $[t, T]$. Under the Black-Scholes model, the homogeneity property of option prices allows the problem to be reduced to the valuation of a standard European option beginning at t, thus, yielding a closed-form analytical solution. In general, the option may be viewed as a compound claim, that is, prior to the grant date the contract represents the right to receive a future option, whilst after t it behaves as a standard European option. Mark Rubinstein's paper, *'Pay Now, Choose Later'* [1991], talks about the valuation of forward-start options where the holder delays the specification of certain contract features until a future date.

To note that this approach could be expanded to stochastic volatility models such as Heston, where valuation is typically performed by conditioning on information available at the grant date and evaluating the expected value of the option under the risk-neutral measure.

1.68 Forward-Start Standard Call

1.68.1 Price

Let $\tau = T - t$, where t denotes the grant date and T the expiration date. The value of a European call with concurrent asset price, S, strike price, E, and time to expiration, τ, is denoted by $C(S, E, \tau)$. From the homogeneity assumption, the value of a forward-start standard at-the-money call option on the grant date, t, is:

$$C(S_t, S_t, \tau) = S_t C(1, 1, \tau).$$

Using the payout assumption, correcting for the loss of dividends over time t, we have the current value of the forward-start option, and by the homogeneity assumption, we obtain:

$$Se^{-qt}C(1, 1, \tau) = e^{-qt}C(S, S, \tau).$$

In other words, the fair value of a forward-start option is simply the current value of e^{-qt} calls which are currently at-the-money, with time to expiration, τ. Thus, via the Black-Scholes model, with underlying asset price and strike, 1, and time to maturity, τ, the fair value of a forward-start standard call option is:

$$C = Se^{-qt}\left(e^{-q\tau}\Phi(d_1) - e^{-r\tau}\Phi(d_2)\right). \tag{92}$$

1.68.2 Sensitivities

Note that $\Gamma = 0$ since C is linear in S.

$$\begin{aligned} \Delta = e^{-qt}\left(e^{-q\tau}\Phi(d_1) - e^{-r\tau}\Phi(d_2)\right), \qquad \mathrm{P} = Se^{-qt}\tau e^{-r\tau}\Phi(d_2) \\ \Theta = -qC + Se^{-qt}\Theta_{BS}(1, 1, \tau), \qquad \mathcal{V} = Se^{-qt}e^{-q\tau}\sqrt{\tau}\,\phi(d_1), \end{aligned} \tag{93}$$

where $\Theta_{BS}(1, 1, \tau)$ is the standard Black-Scholes at-the-money call Theta.

1.69 Forward-Start Standard Put

1.69.1 Price

Similarly:

$$Se^{-qt}P(1, 1, \tau) = e^{-qt}P(S, S, \tau).$$

Thus, the fair value of a forward-start standard put option is:

$$P = Se^{-qt}\left(e^{-r\tau}\Phi(-d_2) - e^{-q\tau}\Phi(-d_1)\right). \tag{94}$$

1.69.2 Sensitivities

Again note that $\Gamma = 0$ since P is linear in S.

$$\begin{aligned} \Delta = e^{-qt}\left(e^{-r\tau}\Phi(-d_2) - e^{-q\tau}\Phi(-d_1)\right), \qquad \mathrm{P} = -Se^{-qt}\tau e^{-r\tau}\Phi(-d_2) \\ \Theta = -qP + Se^{-qt}\Theta_{BS}(1, 1, \tau), \qquad \mathcal{V} = Se^{-qt}e^{-q\tau}\sqrt{\tau}\,\phi(d_1), \end{aligned} \tag{95}$$

where $\Theta_{BS}(1, 1, \tau)$ is the standard Black-Scholes at-the-money put Theta.

FX Options (via the Garman-Kohlhagen Model)

Redefine for sections (1.70) and (1.71), such that:

$$d_1 = \frac{\ln\left(\frac{S}{E}\right) + \left(r_d - r_f - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}},$$

where r_d and r_f represent the domestic and foreign interest rate respectively.

To find the closed-form analytical solution to the price, we solve the Garman-Kohlhagen equation:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r_d - r_f - q)S\frac{\partial V}{\partial S} - rV = 0.$$

1.70 European Call FX Option

1.70.1 Price

The price of the call FX option, C, is:

$$C = Se^{-(r_f+q)T}\Phi\left(d_1\right) - Ee^{-r_d T}\Phi\left(d_2\right). \tag{96}$$

1.70.2 Sensitivities

The Greeks are:

$$\begin{gathered}
\Delta = e^{-(r_f+q)T}\Phi\left(d_1\right), \qquad \Gamma = e^{-(r_f+q)T}\frac{\phi\left(d_1\right)}{S\sigma\sqrt{T}}, \qquad \mathrm{P}_d = Ee^{-r_d T}T\Phi(d_2), \\
\mathrm{P}_f = -Se^{-(r_f+q)T}T\Phi(d_1), \qquad \mathcal{V} = Se^{-(r_f+q)T}\sqrt{T}\,\phi\left(d_1\right), \\
\Theta = -Se^{-(r_f+q)T}\phi(d_1)\frac{\sigma}{2\sqrt{T}} + (r_f+q)Se^{-(r_f+q)T}\Phi(d_1) - r_d Ee^{-r_d T}\Phi(d_2).
\end{gathered} \tag{97}$$

Note that P_d is the sensitivity to the domestic interest rate, r_d, and P_f is the sensitivity to the foreign interest rate, r_f. P_f is mathematically equal to Ψ, the dividend sensitivity.

1.71 European Put FX Option

1.71.1 Price

The price of the put FX option, P, is:

$$P = Ee^{-r_d T}\Phi\left(-d_2\right) - Se^{-(r_f+q)T}\Phi\left(-d_1\right). \tag{98}$$

1.71.2 Sensitivities

$$\begin{gathered}
\Delta = -e^{-(r_f+q)T}\Phi\left(-d_1\right), \qquad \Gamma = e^{-(r_f+q)T}\frac{\phi\left(d_1\right)}{S\sigma\sqrt{T}}, \qquad \mathrm{P}_d = -Ee^{-r_d T}T\Phi(-d_2), \\
\mathrm{P}_f = Se^{-(r_f+q)T}T\Phi(-d_1), \qquad \mathcal{V} = Se^{-(r_f+q)T}\sqrt{T}\,\phi\left(d_1\right), \\
\Theta = -Se^{-(r_f+q)T}\phi(d_1)\frac{\sigma}{2\sqrt{T}} - (r_f+q)Se^{-(r_f+q)T}\Phi(-d_1) + r_d Ee^{-r_d T}\Phi(-d_2).
\end{gathered} \tag{99}$$

Options on Futures (via the Black-76 Model)

Redefine for sections (1.72) and (1.73), such that:

$$d_1 = \frac{\ln\left(\frac{F}{E}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}.$$

1.72 European Call Options on Futures

This is a financial option where the underlying is a future. A futures contract is a financial derivative where the holder must buy or sell an underlying asset at the strike price within a specified time frame. A European call represents that the holder has the right, but not obligation, to buy the future at the strike price at maturity.

To find the closed-form analytical solution to the call price, we change $V(S,t) \to V(F,t)$, where $F = Se^{(r-q)T}$, $\therefore$ let $V = C$ and solve the Black-Scholes equation:

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 F^2 \frac{\partial^2 C}{\partial F^2} - rC = 0.$$

1.72.1 Price

The price of the call option, C, on a future is:

$$C = e^{-rT}\left(F\Phi\left(d_1\right) - E\Phi\left(d_2\right)\right). \tag{100}$$

1.72.2 Sensitivities

In this case, the Greeks are the partial differentials w.r.t. F, thus:

$$\begin{aligned} \Delta &= e^{-rT}\Phi\left(d_1\right), \qquad \Gamma = e^{-rT}\frac{\phi\left(d_1\right)}{F\sigma\sqrt{T}}, \qquad \mathrm{P} = -TC \\ \Theta &= -Fe^{-rT}\phi(d_1)\frac{\sigma}{2\sqrt{T}} + rC, \qquad \mathcal{V} = Fe^{-rT}\sqrt{T}\,\phi\left(d_1\right). \end{aligned} \tag{101}$$

1.73 European Put Options on Futures

A European put represents that the holder has the right, but not obligation, to sell the future at the strike price at maturity.

To find the closed-form analytical solution to the put price, we solve the Black-Scholes equation:

$$\frac{\partial P}{\partial t} + \frac{1}{2}\sigma^2 F^2 \frac{\partial^2 P}{\partial F^2} - rP = 0.$$

1.73.1 Price

The price of the put option, P, on a future is:

$$P = e^{-rT}\left(E\Phi\left(-d_2\right) - F\Phi\left(-d_1\right)\right). \tag{102}$$

1.73.2 Sensitivities

The Greeks are:

$$\begin{aligned} \Delta &= -e^{-rT}\Phi\left(-d_1\right), \qquad \Gamma = e^{-rT}\frac{\phi\left(d_1\right)}{F\sigma\sqrt{T}}, \qquad \mathrm{P} = -TP \\ \Theta &= -Fe^{-rT}\phi(d_1)\frac{\sigma}{2\sqrt{T}} + rP, \qquad \mathcal{V} = Fe^{-rT}\sqrt{T}\,\phi\left(d_1\right). \end{aligned} \tag{103}$$

IN PROGRESS (Further Option Pricing Models)

Implied Volatility (via the Black-Scholes Model)

Minimise the difference between the theoretical price (the Black-Scholes price) and the market price (the mid price calculated from bid-ask spreads in the option chain). Denote x^g as the initial guess for the volatility. Iterate to a closer value for volatility to the true value:

$$x^g_{n+1} = x^g_n - \frac{f\left(x^g_n\right)}{f'\left(x^g_n\right)},$$

where $f\left(x^g_n\right)$ is the difference between the theoretical price (P_{theory}, the Black-Scholes valuation using the initial guess for the volatility) and the actual price (P_{actual}, the mid price), $f'\left(x^g_n\right)$ is the gradient of the difference (the partial derivative of the difference with respect to the volatility). Denote the difference as $\mathcal{L}$, thus:

$$f'\left(x^g_n\right) = \frac{\partial \mathcal{L}}{\partial \sigma} = \mathcal{V},$$

i.e., mathematically equal to Vega. Thus, the Newton-Raphson formula is:

$$\sigma_{n+1} = \sigma_n - \frac{\mathcal{L}\left(\sigma_n\right)}{\mathcal{V}}. \tag{104}$$

To solve for implied volatility iteratively:

Step 1. Make a guess for volatility.
Step 2. Calculate the difference, $\mathcal{L}$.
Step 3. Check if the difference is less than the tolerance, ϵ: $|P_{theory} - P_{actual}| < \epsilon$.
Step 4. Calculate the gradient of the difference, $\frac{\partial \mathcal{L}}{\partial \sigma}$.
Step 5. Update the volatility using the Newton-Raphson formula.
Step 6. Repeat until converged or maximum number of iterations, N_{iter}.

IN PROGRESS (Volatility Modelling Section)

IN PROGRESS (Stochastic Processes Section)

References

Albrecher, Hansjörg, Mayer, Philipp, Schoutens, Wim, and Tistaert, Jurgen (2007). "The Little Heston Trap". In: *Wilmott* 2007 (1), pp. 83–92.

Black, Fischer (1976). "The Pricing of Commodity Contracts". In: *Journal of Financial Economics* 3 (1-2), pp. 167–179. DOI: 10.1016/0304-405X(76)90024-6.

Black, Fischer and Scholes, Myron (1973). "The Pricing of Options and Corporate Liabilities". In: *Journal of Political Economy* 81 (3), pp. 637–654. DOI: 10.1086/260062.

Garman, Mark B. and Kohlhagen, Steven W. (1983). "Foreign Currency Option Values". In: *Journal of International Money and Finance* 2 (3), pp. 231–237. DOI: 10.1016/S0261-5606(83)80001-1.

Geske, Robert (1979). "The Valuation of Compound Options". In: *Journal of Financial Economics* 7.1, pp. 63–81. DOI: 10.1016/0304-405X(79)90022-9.

Hakansson, Nils H. (1976). "The Purchasing Power Fund: A New Kind of Financial Intermediary". In: *Financial Analysts Journal* 32.6, pp. 49–59. DOI: 10.2469/faj.v32.n6.49.

Heston, Steven L. (1993). "A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options". In: *The Review of Financial Studies* 6 (2), pp. 327–343. DOI: 10.1093/rfs/6.2.327.

Kunitomo, Naoto and Ikeda, Masayuki (1992). "Pricing Options with Curved Boundaries". In: *Mathematical Finance* 2 (4), pp. 275–298. DOI: 10.1111/j.1467-9965.1992.tb00033.x.

Lewis, Alan L. (2001). "A Simple Option Formula for General Jump-Diffusion and Other Exponential Lévy Processes". In: *SSRN Electronic Journal*. DOI: 10.2139/ssrn.282110.

Margrabe, William (1978). "The Value of an Option to Exchange One Asset for Another". In: *The Journal of Finance* 33.1, pp. 177–186. DOI: 10.1111/j.1540-6261.1978.tb03397.x.

Merton, Robert C. (1973). "Theory of Rational Option Pricing". In: *Bell Journal of Economics and Management Science* 4.1, pp. 141–183. DOI: 10.2307/3003143.

Reiner, Eric and Rubinstein, Mark (Sept. 1991a). "Breaking Down the Barriers". In: *Risk* 4 (8), pp. 28–35.

– (Oct. 1991b). "Unscrambling the Binary Code". In: *Risk* 4 (20), pp. 75–83.

Rubinstein, Mark (Feb. 1991). "Pay Now, Choose Later". In: *Risk* 4 (2), pp. 44–47.